

Using observational data to improve inference in randomized trials

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Disclaimer: views (and errors) are mine and not those of colleagues, NIH, DHHS, US government



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¹Footnotes are for asides or references, but are fair game for questions

Preventing Vertical HIV Transmission

Standard of Care (SOC):

- Maternal antiretroviral (ARV) therapy
- Short-term daily infant ARV prophylaxis

Transmission still occurs

- Postpartum viral rebound & breastfeeding
 - Still recommended because of survival benefits²

Long-acting injectable (LAI) ARV

- Simplified administration
- Longer efficacy

²Thior et al. *JAMA* 2006

Regulatory Approval for LAI-ARV in Infants

Challenges

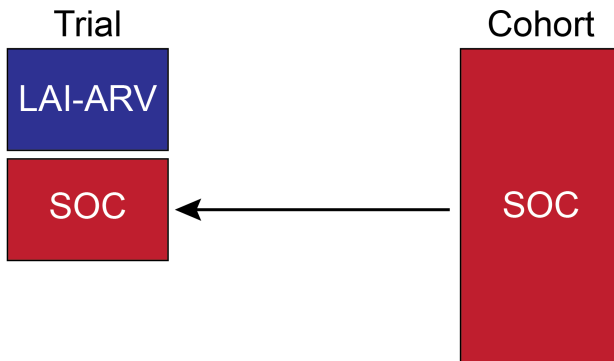
- Extrapolating efficacy from adults unlikely to be acceptable
- Need an active-control trial
 - Low risk of transmission under standard of care³
- Sample sizes difficult to meet, costs

³Around 0.6% in previous trials; Fowler et al. *N Engl J Med* 2016

A Solution

Augmented Trial Design

- Fuse trial and prospective observational cohort



Notation

Y_i^a : potential outcome for unit i under action a

A_i : action for unit i

Y_i : observed outcome for unit i

S_i : population indicator (1: trial 0: cohort)

W_i : baseline covariate related to Y and (maybe) S

$O_i = (S_i, W_i, A_i, Y_i)$

$f_X(x)$: probability density function at x for variable X

$E[\cdot]$: expected value function

$$\tau = \mu_1 - \mu_0 = E[Y^1 \mid S = 1] - E[Y^0 \mid S = 1]$$

Identification

1: Causal Consistency

$$Y_i = Y_i^a \text{ if } a = A_i$$

2.1: Action Exchangeability

$$Y_i^a \perp\!\!\!\perp A_i \mid W_i$$

2.2: Action Positivity

$$\forall w \in \mathcal{W} [0 < \Pr(A = 1 \mid W = w) < 1]$$

3.1: Sampling Exchangeability

$$Y_i^a \perp\!\!\!\perp S_i \mid W_i$$

3.2: Sampling Positivity

$$f_{W|S=1}(w) > 0 \implies f_W(w \mid S = 0) > 0$$

Estimation

Estimating equations are defined as ⁴

The diagram shows the equation $\sum_{i=1}^n \psi(O_i, \hat{\theta}) = 0$ enclosed in a light blue box. Three labels with arrows point to parts of the equation: 'Estimating function' (blue) points to ψ ; 'Estimated parameters' (purple) points to $\hat{\theta}$; and 'Observation i ' (red) points to O_i .

$$\sum_{i=1}^n \psi(O_i, \hat{\theta}) = 0$$

⁴Stefanski & Boos *Am Stat* 2002; Ross et al. *IJE* 2024

Trial-Only Estimator

$$\psi_T(O_i; \theta) = \begin{bmatrix} S_i(1 - A_i)(Y_i - \mu_0) \\ S_i A_i(Y_i - \mu_1) \\ (\mu_1 - \mu_0) - \tau \end{bmatrix}$$

Only trial observations ($S_i = 1$) contribute information

$$\psi_F(O_i; \theta) = \begin{bmatrix} (S_i - \hat{S}_i) \mathbb{W}_i^T \\ \pi_{S_i} (1 - A_i) (Y_i - \hat{Y}_i) \mathbb{X}_i^T \\ S_i (\hat{Y}_i - \mu_0) \\ S_i A_i (Y_i - \mu_1) \\ (\mu_1 - \mu_0) - \tau \end{bmatrix}$$

where

- $\hat{S}_i = \text{expit}(\mathbb{W}_i \alpha^T)$ and $\mathbb{W}_i$ is a design matrix including W
- $\pi_{S_i} = S_i + (1 - S_i) \frac{\hat{S}_i}{1 - \hat{S}_i}$
- $\hat{Y}_i = \text{expit}(\mathbb{X}_i \beta^T)$ and $\mathbb{X}_i$ is a design matrix including W

⁵This estimator is consistent as long as either the sampling model or outcome model is correctly specified

Simulations

Trial

$$\begin{bmatrix} W \sim \text{Bernoulli}(0.\bar{6}) \\ A \sim \text{Bernoulli}(p) \end{bmatrix}$$

Observational Cohort

$$\begin{bmatrix} W \sim \text{Bernoulli}(0.2) \\ A := 0 \end{bmatrix}$$

Outcome model

$$Y \sim \text{Bernoulli}(0.01 + 0.015W(1 - A))$$

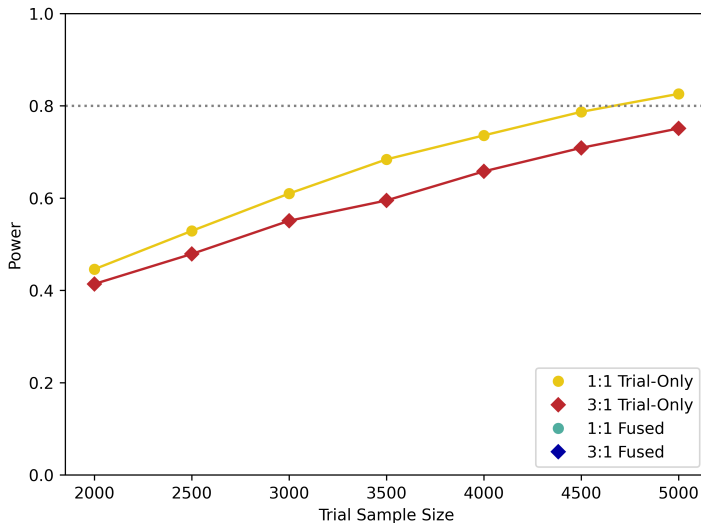
$$\tau = 0.01$$

Assess Power at $\alpha = 0.05$ for

- for 1:1 and 3:1 randomization
- varying trial and observational cohort sizes

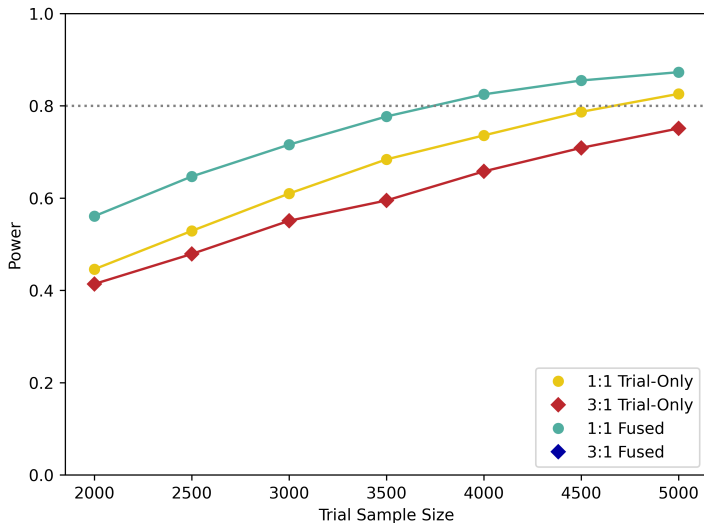
with 5000 data sets

Simulation – Power by Trial Size⁶



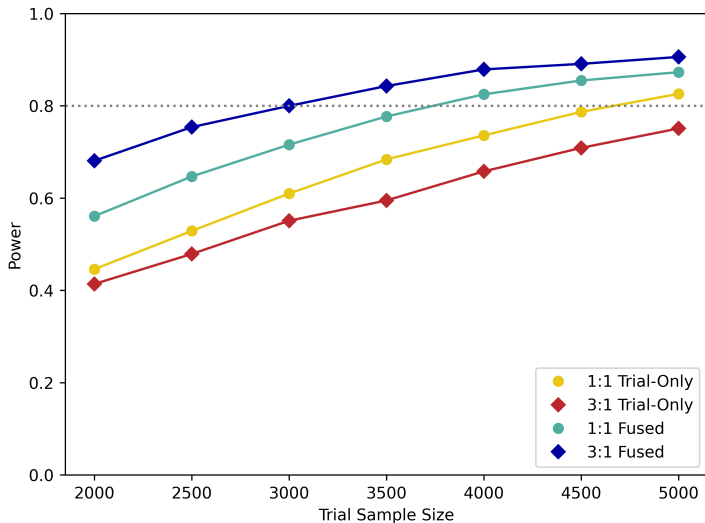
⁶For an observational cohort of 5000

Simulation – Power by Trial Size⁷



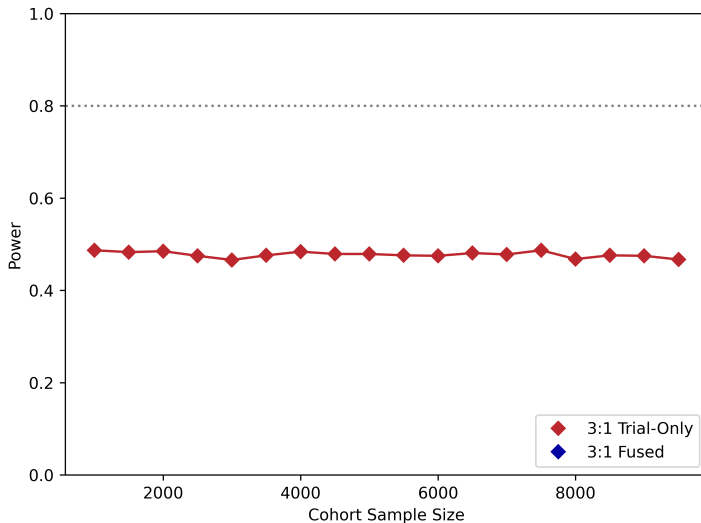
⁷For an observational cohort of 5000

Simulation – Power by Trial Size⁸



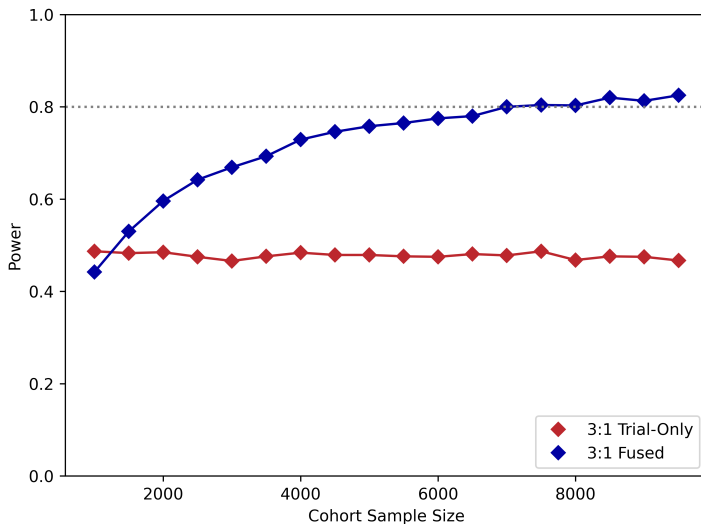
⁸For an observational cohort of 5000

Simulation – Power by Cohort Size⁹



⁹For a trial of 2500

Simulation – Power by Cohort Size¹⁰



¹⁰For a trial of 2500

Conclusions

Trials of LAI-ARV in infants face logistical challenges

- Low event rate under SOC

Fusion estimator

- Improves power over standard trial design
- Tradeoffs
 - Use of two models offers double robustness
 - Less efficient than a g-computation fusion estimator when outcome model is correct

Remaining Challenges

External data may introduce bias

- Violations of 3.1 or 3.2

Design-based mitigation strategies

- Randomize S
- Prospective observational cohort

Analysis-based mitigation strategies

- Randomization-Aware AIPW,¹¹ CV-TMLE¹²

¹¹Gagnon-Bartsch et al. *J Causal Inf* 2023; Karlsson et al. *arXiv:2406.17971*

¹²Dang et al. *arXiv.2210.05802*

Thank you



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Appendix

Performance Metrics

For trial of 2400 and cohort of 5000, operating characteristics were

	Bias	ESE	SER	Coverage	Rejection
Trial-Only	0.000	0.006	0.98	0.93	0.33
Naive Pooling	0.006	0.003	1.00	0.38	0.29
AIPW	0.000	0.004	1.00	0.95	0.77